

Given: Due:	Further Maths (Statistics) HW 26	Name:
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Chapter 6: Chi Square testing (contingency tables)

Q1. A university foreign language department carried out a survey of prospective students to find out which of three languages they were most interested in studying. A random sample of 150 prospective students gave the following results.

		Language		
		French	Spanish	Mandarin
Gender	Male	23	22	20
	Female	38	32	15

A test is carried out at the 1% level of significance to determine whether or not there is an association between gender and choice of language.

- State the null hypothesis for this test. (1)
- Show that the expected frequency for females choosing Spanish is 30.6 (1)
- Calculate the test statistic for this test, stating the expected frequencies you have used. (3)
- State whether or not the null hypothesis is rejected. Justify your answer. (2)
- Explain whether or not the null hypothesis would be rejected if the test was carried out at the 10% level of significance. (1)



Question	Scheme	Marks	AOs																	
1(a)	H ₀ : There is no association between language and gender.	B1	1.2																	
		(1)																		
(b)	$\frac{54 \times 85}{150} = 30.6$ *	B1*cso	1.1b																	
		(1)																		
(c)	<table><tr><th colspan="2" rowspan="2">Expected frequencies</th><th colspan="3">Language</th></tr><tr><th>French</th><th>Spanish</th><th>Mandarin</th></tr><tr><th rowspan="2">Gender</th><th>Male</th><td>26.43...</td><td>23.4</td><td>15.16...</td></tr><tr><th>Female</th><td>34.56...</td><td>[30.6]</td><td>19.83...</td></tr></table>	Expected frequencies		Language			French	Spanish	Mandarin	Gender	Male	26.43...	23.4	15.16...	Female	34.56...	[30.6]	19.83...	M1	2.1
	Expected frequencies			Language																
			French	Spanish	Mandarin															
	Gender	Male	26.43...	23.4	15.16...															
		Female	34.56...	[30.6]	19.83...															
$\chi^2 = \sum \frac{(O - E)^2}{E} = \frac{(23 - 26.43)^2}{26.43} + \dots + \frac{(15 - 19.83)^2}{19.83}$	M1	1.1b																		
awrt <u>3.6/3.7</u>	A1	1.1b																		
	(3)																			
(d)	Degrees of freedom (3 – 1)(2 – 1) → Critical value $\chi^2_{2,0.01} = 9.210$	M1	3.1b																	
	As $\sum \frac{(O - E)^2}{E} < 9.210$, the null hypothesis is not rejected.	A1	2.2b																	
		(2)																		
(e)	Still not rejected since $\sum \frac{(O - E)^2}{E} < \chi^2_{2,0.1} = 4.605$	B1	2.4																	
		(1)																		
(8 marks)																				
Notes																				
(a)	B1 for correct hypothesis in context																			
(b)	B1* for a correct calculation leading to the given answer and no errors seen																			
(c)	M1 for attempt at $\frac{(\text{Row Total})(\text{Column Total})}{(\text{Grand Total})}$ to find expected frequencies																			
	M1 for applying $\sum \frac{(O - E)^2}{E}$ A1 awrt 3.6 or 3.7																			
(d)	M1 for using degrees of freedom to set up a χ^2 model critical value A1 for correct comparison and conclusion																			
(e)	B1 for correct conclusion with supporting reason																			

Q2. John thinks that a person's eye colour is related to their hair colour. He takes a random sample of 600 people and records their eye and hair colours. The results are shown in Table 1.

Table 1

		Hair colour				Total
		Black	Brown	Red	Blonde	
Eye colour	Brown	45	125	15	58	243
	Blue	34	90	10	58	192
	Hazel	20	38	16	26	100
	Green	6	29	7	23	65
	Total	105	282	48	165	600

John carries out a χ^2 test in order to test whether eye colour and hair colour are related. He calculates the expected frequencies shown in Table 2.

Table 2

		Hair colour			
		Black	Brown	Red	Blonde
Eye colour	Brown	42.5	114.2	19.4	66.8
	Blue	33.6	90.2	15.4	52.8
	Hazel	17.5	47	8	27.5
	Green	11.4	30.6	5.2	17.9

(a) Show how the value 47 in Table 2 has been calculated.

(1)

(b) Write down the number of degrees of freedom John should use in this χ^2 test.

(1)

Given that the value of the χ^2 statistic is 20.6, to 3 significant figures,

(c) find the smallest value of α for which the null hypothesis will be rejected at the $\alpha\%$ level of significance.

(1)

(d) Use the data from Table 1 to test at the 5% level of significance whether or not the proportions of people in the population with black, brown, red and blonde hair are in the ratio 2:6:1:3
State your hypotheses clearly.

(9)

(Total 12 marks)

The χ^2 tests were only examined in this one question on this paper and a number of candidates failed to identify the fairly straightforward goodness of fit test in part (d). The first three parts were answered well and most scored full marks although a few misread the tables and thought α was 1 and some gave α as 0.025. In part (d) the common error was to simply calculate the test statistic for a test of association between hair colour and eye colour rather than using the column totals from Table 1 to conduct the goodness of fit test on the hair colour ratios. Those who did attempt the correct test could usually calculate the expected frequencies and often completed the test accurately.



Question Number	Scheme	Marks																									
(a)	$\frac{282 \times 100}{600}$ (Do not accept $282 - 114.2 - 90.2 - 30.6$ (o.e.))	B1 (1)																									
(b)	9	B1 (1)																									
(c)	2.5 or better (Do not accept 0.025)	B1 (1)																									
(d)	H_0: hair colour occurs in the ratio 2:6:1:3 H_1: hair colour does not occur in the ratio 2:6:1:3 <table border="1"> <tr> <td></td><td>black</td><td>brown</td><td>red</td><td>blonde</td></tr> <tr> <td>observed</td><td>105</td><td>282</td><td>48</td><td>165</td></tr> <tr> <td>expected</td><td>100</td><td>300</td><td>50</td><td>150</td></tr> <tr> <td>$\frac{(O_i - E_i)^2}{E_i}$</td><td>0.25</td><td>1.08</td><td>0.08</td><td>1.5</td></tr> <tr> <td>$\frac{O_i^2}{E_i}$</td><td>110.25</td><td>265.08</td><td>46.08</td><td>181.5</td></tr> </table> $\sum \frac{(O_i - E_i)^2}{E_i} = 2.91$ or $\sum \frac{O_i^2}{E_i} - 600 = 602.91 - 600 = 2.91$ (awrt 2.91) $\nu = 3$ cv is 7.815 $[2.91 < 7.815]$ so insufficient evidence to reject H_0 or not significant There is evidence to suggest that hair colour does occur in the given ratio.		black	brown	red	blonde	observed	105	282	48	165	expected	100	300	50	150	$\frac{(O_i - E_i)^2}{E_i}$	0.25	1.08	0.08	1.5	$\frac{O_i^2}{E_i}$	110.25	265.08	46.08	181.5	B1 B1 B1 expected M1 A1 A1 B1 B1 dM1 A1 (9) [Total 12]
	black	brown	red	blonde																							
observed	105	282	48	165																							
expected	100	300	50	150																							
$\frac{(O_i - E_i)^2}{E_i}$	0.25	1.08	0.08	1.5																							
$\frac{O_i^2}{E_i}$	110.25	265.08	46.08	181.5																							
	Notes																										
(d)	1st B1 for both hypotheses. Must mention hair colour and ratio e.g. "hair colour in the given ratio" Allow use of ditto 2nd B1 for all 4 correct expected frequencies 1st M1 for at least 2 correct calculations from 3rd or 4th row 1st A1 for all correct calculations to at least 3sf if row 4 If awrt 2.91 is seen with no incorrect working award B1M1A1A1 2nd dM1 Dep on 1st M1 for a correct statement linking their test statistic and their cv (cv > 3.5) 3rd A1 for a correct comment in context - must mention "hair colour" and "ratios" or "model" e.g. "There is evidence of to support the given model" No follow through If hypotheses are the wrong way round score A0.																										

Q3. A leisure club offers a choice of one of three activities to its 150 members on a Tuesday evening. The manager believes that there may be an association between the choice of activity and the age of the member and collected the following data.

(a) Write down suitable hypotheses for a test of the manager's belief.

(1)

The manager calculated expected frequencies to use in the test.

Activity Age a years	Badminton	Bowls	Snooker
$a < 20$	9	3	3
$20 \leq a < 40$	10	10	14
$40 \leq a < 50$	16	15	5
$50 \leq a < 60$	15	13	11
$a \geq 60$	4	19	3

(b) Calculate the expected frequency of members aged 60 or over who choose snooker, used by the manager.

(1)

(c) Explain why there are 6 degrees of freedom used in this test.

(2)



The test statistic used to test the manager's belief is 19.583

(d) Using a 5% level of significance, complete the test of the manager's belief.

(2)

(Total for question = 6 marks)

Nearly all the candidates knew the hypotheses were about association or independence and mentioned the context but a surprising number gave H_0 and H_1 the wrong way around. Part (b) was answered very well and the critical value in part (d) was usually quoted correctly with these two marks being scored by the vast majority of the candidates. The explanation in part (c) was not always answered thoroughly enough. We were looking for candidates to realise that some of the rows would need amalgamating because some expected frequencies were less than 5. Some calculated all of the expected frequencies but their calculation in part (b) should have steered them towards the top right-hand cell which had an expected frequency of 3.6 and therefore meant that some amalgamation was required. Those who did identify this value usually realised that amalgamating the first two rows would deal with the small expected frequencies and the ensuing 4x3 table would lead to 6 degrees of freedom. Few candidates calculated the 3.6 (though one mark was available for mentioning the combining of rows leading to the 4x3 table) and some confused rows and columns or just mentioned vaguely combining cells. A common error was to think that amalgamation of cells was required because of the observed frequencies being less than 5 and this sometimes led to combining snooker and bowls or the first and last rows. Others simply thought that the 6 came from a calculation of $8 - 2$ presumably confusing with the calculations required for goodness of fit tests. In the final part only a small minority did not know how to use the tables and gave the value 1.635 and those who had their hypotheses the correct way round in part (a) were usually able to give an appropriate conclusion either in terms of there being evidence of association between age and choice of activity or, more simply, stating that there was evidence to support the manager's belief.

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	Scheme	Marks	AO
(a)	H_0 : There is no association between age and activity H_1 : There is an association between age and activity	B1 (1)	1.2
(b)	$\frac{26 \times 36}{150} = 6.24$	B1 (1)	1.1b
(c)	Since expected value in $a < 20$ and snooker = $3.6 < 5$ we amalgamate two rows Table is now 4×3 so degrees of freedom is $(4-1) \times (3-1) = 6$ (*)	B1 B1* (2)	2.4 1.1b
(d)	Critical value $\chi^2_6(5\%) = 12.592$ [Significant result]: so there is evidence to support manager's belief	B1 B1ft (2)	1.1b 2.2b
		(6 marks)	

Notes

(a)	B1 for both hypotheses in terms of "association" or "independence". Must mention age and activity (or sport). [Use of "relationship" or "link" here is B0 but allow for last B1ft]
(b)	B1 for 6.24
(c)	1 st B1 for a reason to get a 4×3 table based on amalgamation of rows Must mention $a < 20$ and snooker and see 3.6 and be combining <u>rows</u> (not columns) 2 nd B1* for 6 degrees of freedom clearly coming from a 4×3 table formed from amalgamation of <u>rows</u> . [$8 - 2 = 6$ is B0]
(d)	1 st B1 for correct critical value (allow 12.6 or 12.59 or awrt 12.592) NB p -value = 0.0032839... so allow awrt 0.00328 2 nd B1ft for a correct comparison and conclusion (ft their cv) [<u>Independent of hypotheses</u>] e.g. there is an "association" or "relationship" or "link" between age and activity is OK BUT there is a "correlation" between age and activity is B0 Do not accept contradictory contextual statements e.g. "manager's belief supported, there is no association between age and activity"